

Module 6: Interest and Loans — Answer Key

Prequiz

- $$A = 1500(1 + 0.04 \cdot 3) = 1500(1.12) = \boxed{\$1680}.$$

Yearly interest is $0.04(1500) = \boxed{\$60}$.
- $$A = 1000(1.05)^4 = \boxed{\$1215.51}.$$

The dollar interest grows because each year's interest is computed from the current balance.
- Doubling time $\approx 72/6 = \boxed{12 \text{ years}}$. Four times the original amount requires about two doublings, or $\boxed{24 \text{ years}}$.
- Monthly rate:

$$0.09/12 = \boxed{0.0075}.$$

Interest:

$$3000(0.0075) = \boxed{\$22.50}.$$

Principal reduction:

$$150 - 22.50 = \boxed{\$127.50}.$$

New balance:

$$3000 + 22.50 - 150 = \boxed{\$2872.50}.$$
- First-month interest is the **same** because the starting balance and rate are the same. Loan B, with the \$250 payment, will generally be paid off sooner because each payment removes more principal, which also reduces later interest.

Matching quiz

- $$A = 2000(1 + 0.03 \cdot 5) = \boxed{\$2300}.$$

Yearly increase = $0.03(2000) = \boxed{\$60}$.
- $$500(1.08)^3 = \boxed{\$629.86}.$$
- Doubling time = $72/8 = \boxed{9}$ years; two doublings $\approx \boxed{18}$ years.
- Monthly rate = $0.12/12 = \boxed{0.01}$. Interest = $6000(0.01) = \boxed{\$60}$. Principal reduction = $200 - 60 = \boxed{\$140}$. New balance = $6000 + 60 - 200 = \boxed{\$5860}$.
- First-month interest is the same. The \$250 payment removes \$100 more principal than the \$150 payment because the interest charge is identical.